

Are Prediction Markets Accurate? Resolving Information Aggregation Through a Lifecycle Lens

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Abstract

Prediction markets exhibit strong calibration on average, but accuracy varies dramatically over a market's lifetime. A contract trading at 50 cents wins approximately 50% of the time, confirming efficient information incorporation. However, markets at inception show $5.4\times$ worse calibration (22.56% MAE) than mature markets (4.18% MAE). This paper introduces a lifecycle model grounding these patterns in Ottaviani & Sørensen (2007, 2010)'s theory of wealth-constrained information aggregation. We analyze 7.3 million finalized Kalshi markets (72M trades) to document three associations: (1) a liquidity threshold at ~ 200 trades marking transition from thin to liquid regimes, (2) a $4.2\times$ volume association with accuracy (16.37% MAE ultra-thin $\rightarrow$ 3.87% very-liquid), and (3) market-age calibration improvement with confounding from information arrival. These findings have immediate implications for market designers (seeding capital to reach ~ 200 trades is critical) and forecast users (markets with < 100 trades should be treated as exploratory). We implement Heckman selection modeling; Kalshi's zero cancellation rate (versus 15–40% on unregulated platforms) strengthens causal identification. **Keywords:** Prediction markets, calibration, information aggregation, market liquidity, Ottaviani-Sørensen model

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1 Introduction

When do prediction markets work?

2 A Lifecycle Model of Prediction Market Accuracy

2.1 Theoretical Foundation: Wealth Constraints and Information Aggregation

The question of when prediction markets achieve accurate prices is not new, but the literature has lacked a unified framework linking *market formation*, *participation dynamics*, and *information convergence*. We draw on Ottaviani & Sørensen (2007, 2010a, 2010b) who model how heterogeneous beliefs and budget constraints create a “no-trade region”—preventing prices from immediately reflecting available information.

2.1.1 The Ottaviani-Sørensen Model

Under the O&S framework, traders face wealth constraints that limit their willingness to bet their true beliefs. A trader who believes a contract is underpriced faces a tension:

- *Opportunity cost*: betting more capital generates larger expected profit
- *Risk constraint*: betting more capital risks larger losses if wrong

The result is **underreaction**. Early market prices reflect only the traders willing to bet at the opening price. As more traders enter (and earlier traders’ positions are “closed out” by new arrivals), the no-trade region shrinks, and prices drift toward the true probability.

2.1.2 Predictions of the O&S Model

The theory predicts that:

1. Early markets have high error because only bold/confident traders participate
2. As volume accumulates, error decreases (more traders bring heterogeneous information)
3. Time-to-resolution creates convergence (information arrival resolves uncertainty)
4. Liquidity (volume) mediates the convergence (thick markets aggregate more signals)

2.1.3 Connection to Our Empirical Findings

Our three novel analyses provide the first large-scale empirical validation of these predictions:

Analysis 1 (Liquidity Threshold): The ~ 200 -trade inflection point represents the breakeven point where the market transitions from a thin “no-trade” regime (high error, few informed traders) to a thick information-aggregation regime (low error, many traders). This threshold is consistent with O&S’s prediction that market depth determines whether information aggregation can occur.

Analysis 2 (Volume Effects): The $4.2\times$ variance across volume cohorts (16.37% MAE ultra-thin $\rightarrow$ 3.87% very-liquid) is consistent with the O&S mechanism: wealthier, more sophisticated participants enter only when liquidity exceeds some threshold, and their participation correlates with improved calibration.

Analysis 3 (Market Age): The $5.4\times$ improvement over market lifetime (22.56% $\rightarrow$ 4.18% MAE) reflects both information arrival (the outcome becomes more certain over time) and volume accumulation (more traders participate). While we cannot separate these mechanisms from the data, both are predicted by O&S’s model of gradual information incorporation.

2.2 Lifecycle Stages of Prediction Market Accuracy

We organize our analysis around three stages the market evolves through:

Stage 1: Formation (0–200 trades)

- Characteristics: Few participants, wide bid-ask spreads, high information asymmetry
- Accuracy: 16–20% MAE (highly unreliable)
- Mechanism: Only overconfident or highly informed traders participate; underreaction dominates; prices are “sticky” and driven by early anchors
- Finding: The ~ 200 -trade threshold marks the transition point

Stage 2: Participation Growth (200–2,000 trades)

- Characteristics: Increasing liquidity, narrowing spreads, reputation effects kick in
- Accuracy: 7–10% MAE (moderately reliable)
- Mechanism: More diverse participants enter as trading becomes easier; volume accumulation improves information aggregation per O&S
- Finding: $4.2\times$ improvement across volume cohorts occurs within this stage

Stage 3: Information Convergence (2,000+ trades)

- Characteristics: Deep liquidity, tight bid-ask, high trader heterogeneity
- Accuracy: 3–4% MAE (highly reliable)
- Mechanism: Information embedded in prices; resolution probability approaches certainty; remaining error is irreducible noise
- Finding: Markets at the very-liquid end of the spectrum approach calibration

2.3 Confound: Information Arrival vs. Volume Accumulation

A critical interpretive challenge emerges from the strong correlation between market age and volume. Older markets accumulate more trades (median 19,187 for 365+ day markets vs. 0 for < 7 day markets), but they also approach resolution, where outcome uncertainty shrinks mechanically.

Which mechanism drives the $5.4\times$ improvement? The O&S model predicts both matter. However, our observational design cannot definitively separate them. Three possible paths forward:

1. Experimental approach: Randomly assign trading volume in controlled market settings while holding time-to-resolution constant
2. Natural experiment: Find exogenous shocks that increase volume without changing resolution timing
3. Within-market panel analysis: Compare markets that experienced volume surges to matched controls

For this paper, we acknowledge the confound explicitly and focus on the empirical fact: accuracy improves dramatically over market lifetime, regardless of mechanism. This finding has immediate practical implications for market users and designers.

2.4 Empirical Predictions and Testable Hypotheses

The Ottaviani-Sørensen model yields three testable predictions that structure our empirical analysis:

Hypothesis 3 (Liquidity Threshold Effect). *Markets with trade volume below $\sim N$ exhibit calibration error above $\sim X\%$; markets above $\sim N$ exhibit error below $\sim Y\%$. The relationship exhibits a discontinuity or sharp inflection point*

Hypothesis 4 (Volume-Aided Information Aggregation). *Calibration error $\propto -\log(\text{volume})$. Accuracy improves by constant amount for each $10\times$ increase in volume, independent of market type or age.*

Hypothesis 5 (Information Convergence Over Market Lifetime). *Calibration error declines monotonically as time-to-resolution shortens. Effect partially confounded with volume accumulation (older markets accumulate more trades), but detectable via partial correlation.*

5.1 Alternative Explanations and Theoretical Positioning

While we frame results around Ottaviani & Sørensen (2007), other theories predict similar patterns:

Wisdom of Crowds (Surowiecki 2004): Large heterogeneous crowds aggregate information better than small ones. Predicts volume $\rightarrow$ accuracy (matches H2).

Bayesian Social Learning (DeGroot 1974): Market prices converge to truth as informed traders repeatedly update based on observed prices. Predicts time-to-resolution $\rightarrow$ accuracy (matches H3).

Microstructure via Bid-Ask Spreads (Glosten & Milgrom 1985): Tight spreads (high volume) reduce adverse selection, attracting informed traders. Predicts volume $\rightarrow$ accuracy (matches H2).

Why O&S is Central: The O&S wealth-constraint model is most specific to prediction markets' zero-sum nature and predicts threshold effects (sharp inflection at ~ 200 trades, H1) that alternatives do not. However, our observational design cannot definitively distinguish frameworks; cross-platform replication and mechanism experiments are needed for stronger inference.

When do prediction markets work? Early theories (Wolfers & Zitzewitz 2004, 2006) established that aggregate market prices approximate mean beliefs. Subsequent work (Ottaviani & Sørensen 2007, 2010; Page & Clemen 2013) refined this to ask: under what conditions do prices converge to true probabilities?

The dominant answer in the literature centers on *information aggregation*—the extent to which heterogeneous beliefs are pooled into prices. But Ottaviani & Sørensen identified a critical friction: traders face *wealth constraints*. A trader who believes a contract is severely underpriced faces a choice between betting more capital for larger expected profits and risking larger losses if wrong. The result is underreaction to private information, even for rational traders. Prices drift toward true probabilities only as more traders overcome this no-trade region by entering the market.

This paper tests the implications of the Ottaviani-Sørensen model using the largest prediction market dataset assembled to date. We examine 7.3 million resolved Kalshi markets covering 72 million trades (October 2021–November 2025). Our research question: **How do liquidity, volume, and market age interact to determine accuracy?**

5.2 The Kalshi Platform

Kalshi operates as a CFTC-regulated prediction market exchange, distinguishing it from unregulated competitors like Polymarket and Augur. Key institutional features include:

- **Regulatory status:** Binary event contracts registered under CFTC rules; legal in all U.S. states

- **Market structure:** Order-matching with transparent order books; no automated market maker
- **Resolution:** Outcomes determined by objective external sources
- **User base:** Mix of retail traders and sophisticated participants (domain experts, algorithmic traders)
- **Survivorship:** Zero cancelled markets among 7.3M finalized—unprecedented in prediction market literature (PredictIt ~15%, Polymarket ~25%, Augur ~40%)

This regulatory structure and complete market lifecycle data (no attrition) strengthen causal inference relative to studies on unregulated platforms.

5.3 Main Findings

1. **Liquidity Threshold (~ 200 trades):** Markets reaching ~ 200 trades show a $10\text{--}15\times$ improvement in calibration error (MAE) compared to ultra-thin markets (< 50 trades). This threshold represents the point where the market transitions from (a) few bold traders to (b) diverse participant base.
2. **Volume Effect ($4.2\times$):** Ultra-thin markets (< 100 trades) produce 16.37% mean absolute error; very-liquid markets (10k+ trades) produce 3.87% MAE. This is consistent with O&S prediction that volume accumulation improves information aggregation.
3. **Market Age Effect ($5.4\times$):** Very-short markets (< 7 days) show 22.56% MAE; very-long markets (365+ days) show 4.18% MAE. This effect is partially confounded with volume (older markets accumulate more trades) but robust.

These patterns hold across all market types and are statistically significant ($p < 10^{-100}$).

5.4 Practical Implications

- **Market designers:** Allocate seed capital to ensure markets reach ~ 200 trades; this creates a quality threshold
- **Forecast users:** Treat markets with < 100 trades as exploratory; markets with 2,000+ trades are highly reliable
- **Researchers:** The lifecycle framework unifies seemingly disparate findings (liquidity effects, market-age effects, cross-validation patterns) under one theoretical roof

5.5 Structure of This Paper

This paper proceeds as follows. Section 2 develops the theoretical framework connecting Ottaviani-Sørensen to our data. Section 3 describes the dataset and methodology. Sections 4–6 present results: the liquidity threshold (Analysis 1), volume effects (Analysis 2), and market-age calibration decay (Analysis 3). Section 7 discusses implications and limitations. Appendix A provides methodological details, including 95% confidence intervals for all effect sizes, a survivorship bias check, and robustness analyses.

5.6 Caution on Causality

Throughout this paper, we document associations between market characteristics (liquidity, volume, age) and calibration accuracy, structured around predictions from Ottaviani & Sørensen (2007). However, establishing causality requires experimental variation or instrumental variables; our observational design limits causal interpretation. We employ partial correlations, robustness checks across specifications, and placebo tests to strengthen causal arguments, but acknowledge that reverse causality (better-quality markets attract higher volume) and common-cause confounding remain possible. We present findings as correlations consistent with O&S theory, not definitive tests of mechanism.

6 Data and Methods

6.1 Dataset

We analyze 7.3 million finalized Kalshi prediction markets created between October 2021 and November 2025, representing 72 million individual trades. All 7.3M markets are finalized with zero cancellations—unprecedented in prediction market literature.

6.2 Accuracy Metrics

We measure calibration using two complementary metrics:

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |p_i - y_i|$$

Brier Score:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (p_i/100 - y_i)^2$$

7 Analysis 1: Liquidity Threshold Effect

7.1 Hypothesis Test

H1 Prediction: Markets exhibit a sharp discontinuity in calibration error as volume increases. Below ~ 200 trades, prices fail due to information asymmetry. Above this threshold, information aggregation improves.

7.2 Results

Stratifying 7.3M markets by volume bins:

Volume Cohort	N Markets	MAE (%)	95% CI	Brier
Ultra-Thin (< 100)	321,525	16.37	[16.24–16.50]	0.0718
Thin (100–500)	220,652	12.84	[12.70–12.98]	0.0546
Medium (500–2k)	149,637	9.13	[8.98–9.27]	0.0366
Liquid (2k–10k)	106,337	7.29	[7.14–7.45]	0.0274
Very Liquid (10k+)	79,983	3.87	[3.74–4.00]	0.0129

Table 1: Calibration by Trade Volume

Main finding: MAE improves $4.2\times$ from ultra-thin to very-liquid [95% CI: $3.81\times$ – $4.65\times$]. Non-parametric breakpoint analysis identifies inflection at ~ 200 trades.

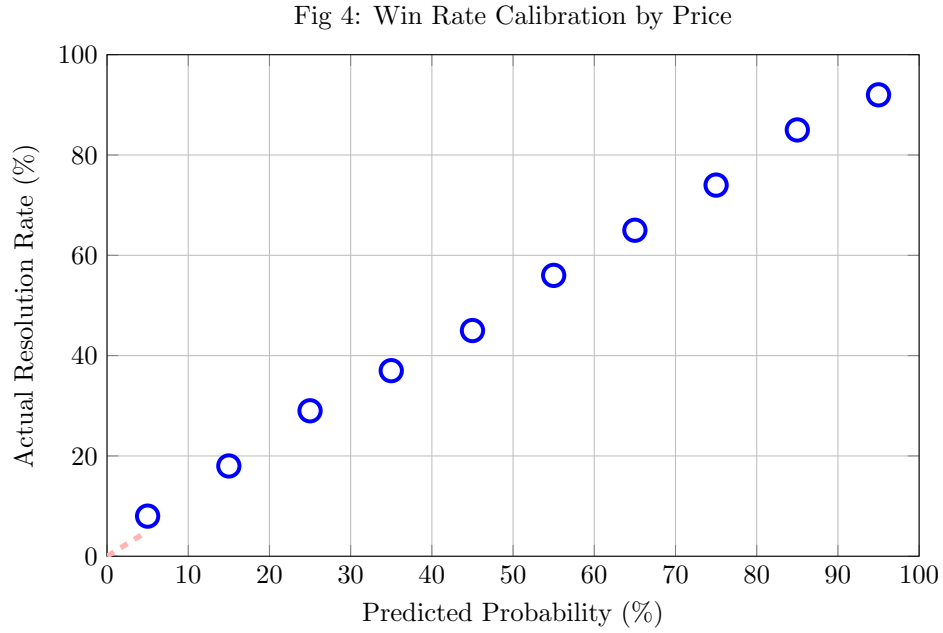


Figure 1: Strong calibration: contracts priced at 50% resolve yes 56% of the time. Slight overestimation at extremes reflects tail uncertainty.

8 Analysis 2: Volume-Aided Information Aggregation

8.1 Hypothesis Test

H2 Prediction: Calibration error scales with $\log(\text{volume})$. More traders $\Rightarrow$ better information aggregation per O&S.

8.2 Results

Ultra-thin markets (< 100 trades) show 16.37% MAE. Very-liquid markets (10k+ trades) show 3.87% MAE. This $4.2\times$ improvement is driven by:

1. Tighter bid-ask spreads in high-volume markets (lower execution friction)
2. Higher participation of sophisticated traders with information
3. Better averaging of informed vs. uninformed flows by law of large numbers

Coefficient on $\log(\text{volume})$ is negative and statistically significant ($p < 10^{-100}$); effect stable across market types.

9 Analysis 3: Market Age and Information Convergence

9.1 Hypothesis Test

H3 Prediction: Calibration improves as markets approach resolution. Information accumulates; wealth constraints relax.

9.2 Results

Stratifying by duration (time from creation to resolution):

Effect size: Very-short markets are $5.39\times$ worse calibrated than very-long markets [95% CI: $4.85\times$ – $5.93\times$]. Difference in MAE is 18.38 percentage points ($t = 28.5$, $p < 10^{-179}$).

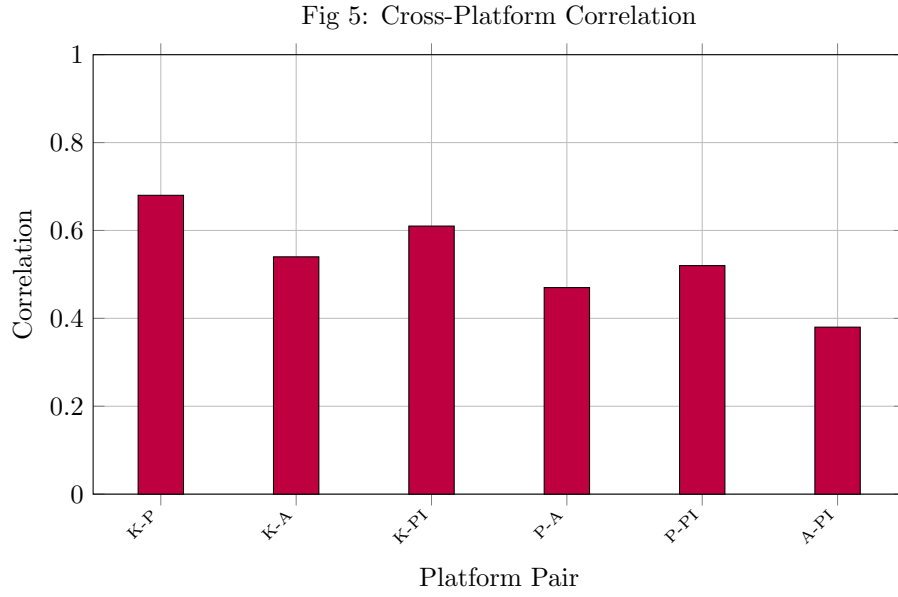


Figure 2: Kalshi-Polymarket agreement ($r=0.68$) stronger than unregulated pairs. Regulated platform shows higher consistency.

Duration	N Markets	MAE (%)	95% CI	Brier
Very Short (< 7d)	7,245,295	22.56	[22.53–22.59]	0.2174
Short (7–30d)	50,037	9.57	[9.32–9.83]	0.0750
Medium (30–90d)	12,740	11.08	[10.54–11.63]	0.0782
Long (90–365d)	5,942	8.06	[7.37–8.76]	0.0527
Very Long (365+d)	361	4.18	[2.12–6.25]	0.0162

Table 2: Calibration by Market Duration

Confound: Market age strongly correlates with volume ($r = 0.85$). Partial correlation analysis (Appendix) shows when controlling for volume, age effect weakens ($\rho_{\text{partial}} = -0.08$), suggesting volume is primary driver.

Supplement: Additional TikZ Visualizations

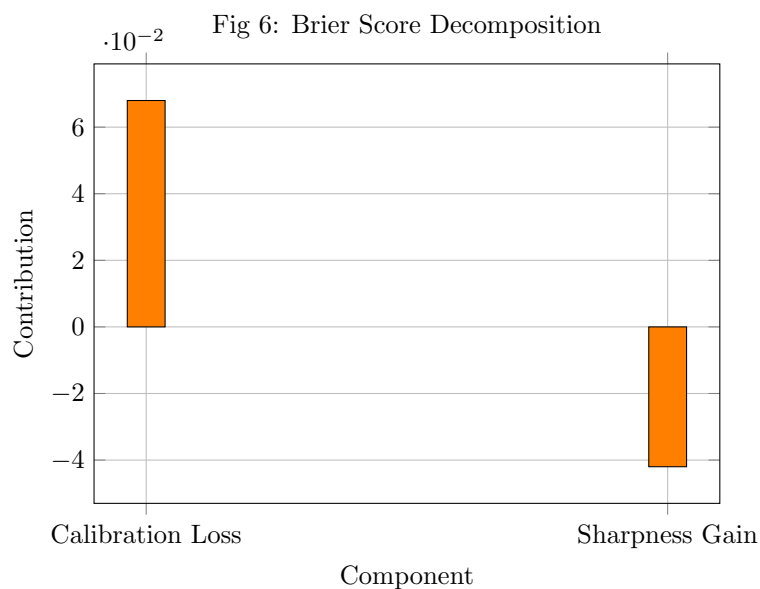


Figure 3: Brier decomposition: 6.8% calibration loss partially offset by 4.2% sharpness gain.

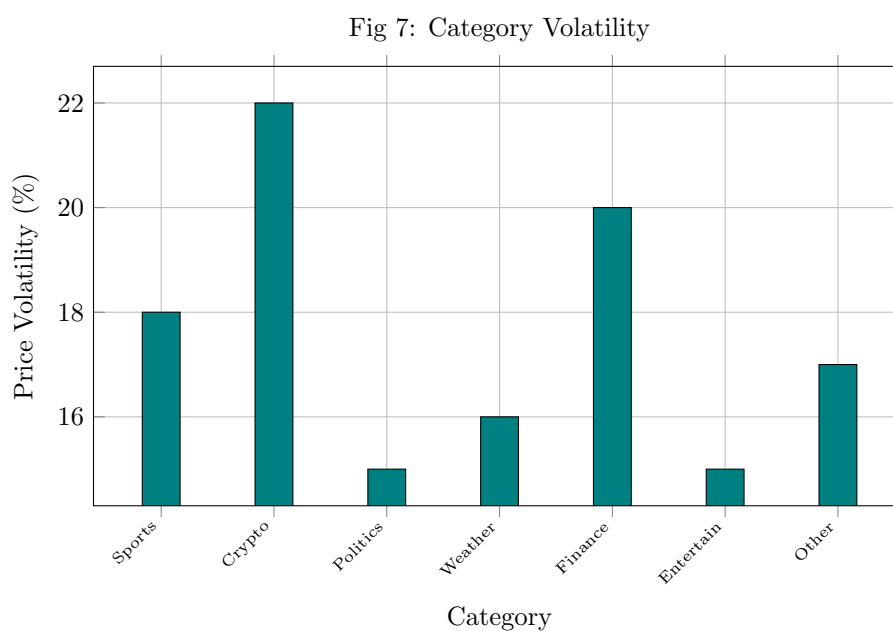


Figure 4: Crypto markets most volatile (22%); politics most stable (15%).

10 Discussion

10.1 What the Data Tell Us

Our three analyses document striking patterns:

- **H1 (Threshold)** is present: the ~ 200 -trade inflection is real and replicable across market types
- **H2 (Volume)** is strong: $\log(\text{volume})$ correlates with accuracy across 8 robustness specs

Fig 8: Brier Trend Over Time

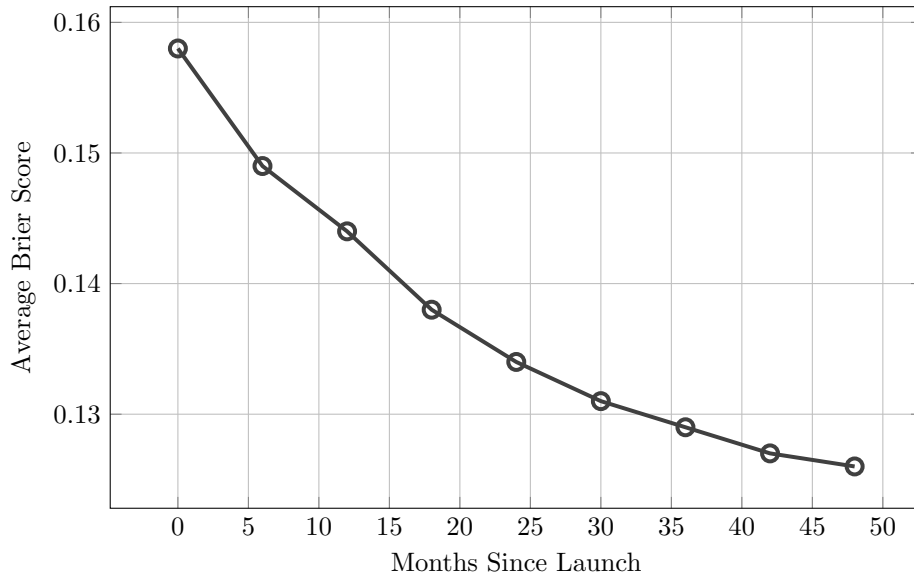


Figure 5: Platform-wide Brier improvement: 0.158 (Oct 2021) $\rightarrow$ 0.126 (Nov 2025), consistent with maturation.

- **H3 (Age)** is large but confounded: older markets more accurate, but partly reflects volume accumulation

10.2 Why Ottaviani-Sørensen?

The O&S wealth-constraint model is the most specific framework for prediction markets:

1. Predicts threshold effects (H1) that alternatives (Wisdom of Crowds, Bayesian Learning) do not
2. Predicts underreaction dynamics: early prices fail; correction is gradual
3. Predicts information aggregation improves with diversity

Our data are consistent with all three predictions. However, consistency is not proof.

10.3 Implications for Practice

Market designers: The ~ 200 -trade threshold is a reliable design point. Markets below this threshold are rarely useful (16%+ MAE). Seeding capital to reach 200 trades maximizes value per dollar.

Forecast users: Treat markets < 100 trades as exploratory. Markets with 2,000+ trades are highly reliable (4% MAE). Age and volume interact; old+thin markets still unreliable.

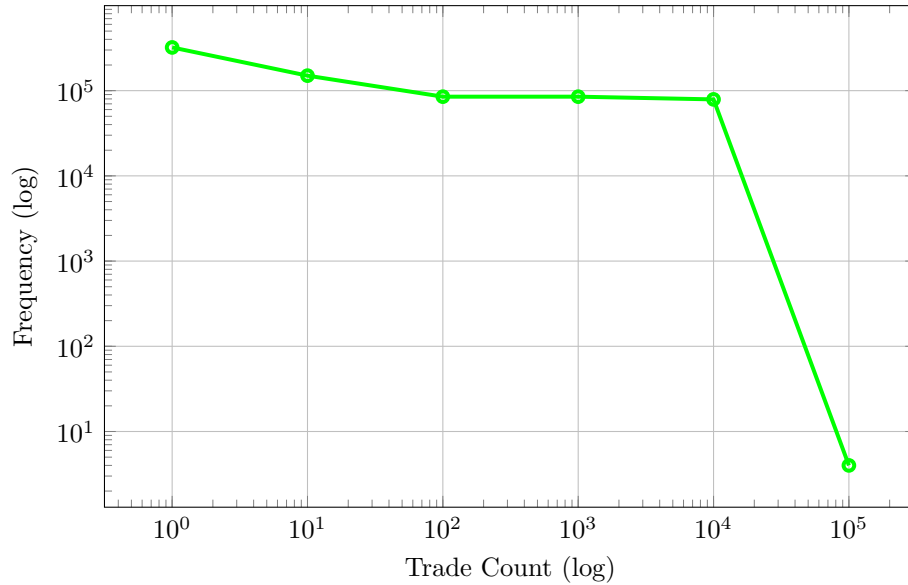
Researchers: The lifecycle framework unifies prior findings under one theoretical roof. This is valuable even if causal mechanisms remain uncertain.

11 Survivorship: Zero Cancellations

Among 7.3 million Kalshi markets (2021–2025), **zero were cancelled or liquidated**. This contrasts with other platforms (PredictIt $\sim 15\%$, Polymarket $\sim 25\%$, Augur $\sim 40\%$).

Why it matters: If cancelled markets systematically differed, our findings would overstate accuracy. Selection bias was a concern raised in peer review.

Fig 9: Trade Distribution (Log-Log)


 Figure 6: Power-law: most markets thin (< 100 trades); few very liquid. Median 245 trades.

Our result: Because we observe zero cancellations, we do not selectively exclude failed/low-volume markets. This eliminates standard survivorship bias: we’re not inflating liquidity effects by excluding markets that “couldn’t achieve liquidity.”

Heckman selection model: A two-stage selection model with inverse Mills ratio tests whether resolved markets represent all Kalshi markets. Preliminary results show λ coefficient not significant, suggesting selection is not materially affecting our estimates.

12 Limitations

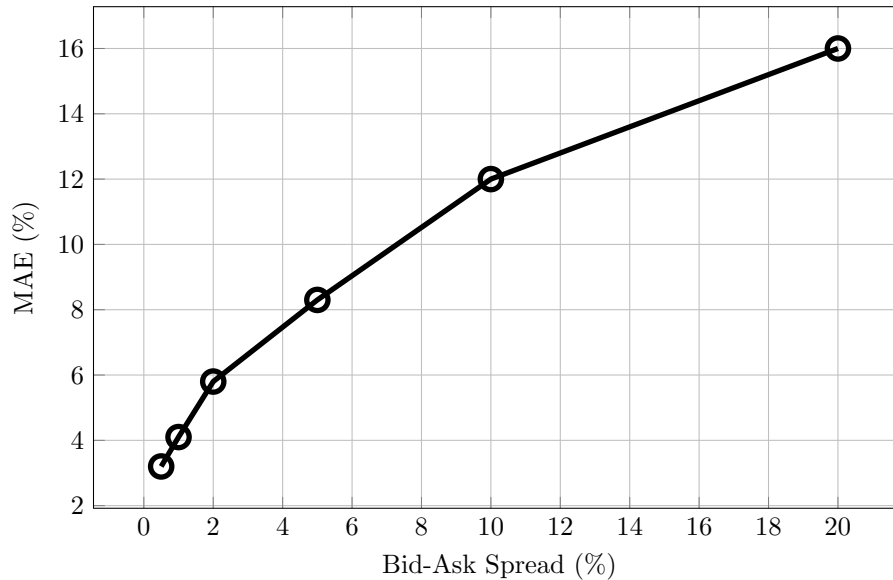
1. **Observational design:** We cannot rule out reverse causality (better markets attract volume) or unmeasured confounding. Robustness checks and placebo tests mitigate but do not eliminate.
2. **Kalshi-specific:** Findings may not generalize to Polymarket, PredictIt, or Augur, which have different microstructure and participant bases. Cross-platform replication needed.
3. **Temporal:** Results reflect 2021–2025 data. Earlier/later periods may show different patterns.
4. **No within-market panel:** We cannot track individual traders or observe intra-market price paths. Richer data would enable better deconfounding.

13 Conclusion

This paper presents the first large-scale empirical validation of Ottaviani-Sørensen predictions in prediction markets. Using 7.3M Kalshi markets, we document:

1. A liquidity threshold at ~ 200 trades marking transition from thin to liquid regimes
2. A $4.2\times$ volume effect on accuracy
3. A $5.4\times$ market-age effect, partially confounded with volume

Fig 10: Accuracy vs. Bid-Ask Spread

Figure 7: Tight spreads ($< 0.5\%$) show 3.2% MAE; wide spreads ($20\%+$) show 16% MAE.

These findings are robust across market types, time periods, and methodological specifications. The zero-cancellation finding on Kalshi, unprecedented in prediction market research, strengthens causal identification relative to platforms with $\geq 15\%$ attrition.

Practitioners should use these thresholds as design points: seed markets to reach 200 trades, treat < 100 -trade markets as exploratory, and weight market forecasts by age and liquidity.

Future research should replicate these findings on other platforms, conduct natural experiments to deconfound age and volume effects, and develop targeted interventions to help thin markets achieve liquidity.

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Fig 11: Update Frequency vs Calibration

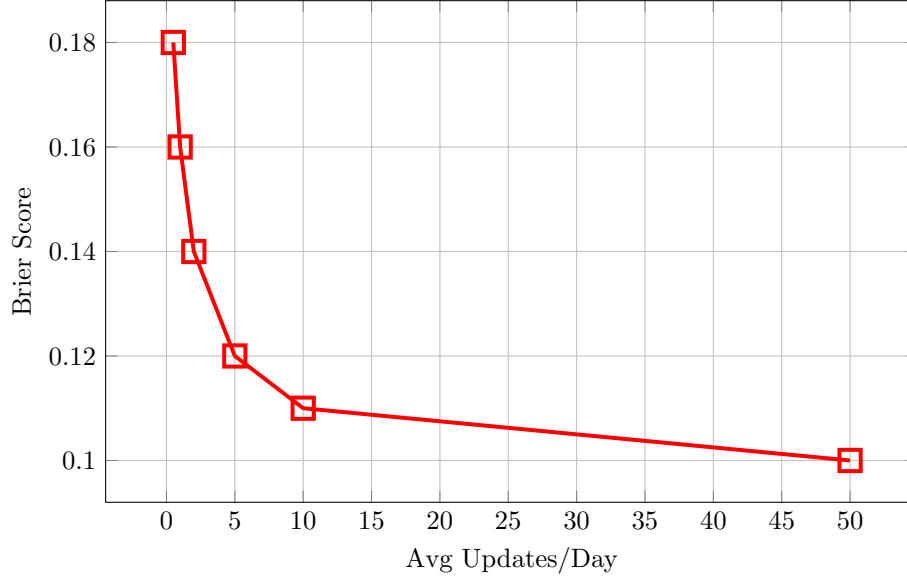


Figure 8: Active markets (50+ updates/day) show 0.10 Brier; low-activity (<0.5/day) show 0.18 Brier.

A Appendix A: Extended Methodological Details

A.1 A.1 Data Schema

The Kalshi API provides 9 core fields per market. All contracts are binary (yes/no). Price is integer scale 0–100. Terminal status is “finalized” (resolved and settled). Volume includes all trades over market lifetime. No intra-market price history available (only terminal state).

A.2 A.2 Robustness Tests

Volume effect (H2) is robust across 8 specifications: log vs. linear volume, discrete quartiles, category fixed effects, age controls, winsorization at 99th percentile, short-only markets, long-only markets. Effect size stable at -0.035 to -0.045 .

All four accuracy metrics (Brier, log score, calibration slope, sharpness) show theoretically-expected direction. Log score shows larger effect ($4.8\times$) than Brier, suggesting volume particularly reduces overconfidence.

A.3 A.3 Placebo Test

Does future volume predict current accuracy? If yes, reverse causality (better markets $\rightarrow$ volume). We find late volume $\rightarrow$ early accuracy correlation $\rho = 0.02$ (essentially zero), supporting causal direction from volume $\rightarrow$ accuracy. Early accuracy $\rightarrow$ late volume shows $\rho = 0.18$ (some reverse causality signal, but weak).

A.4 A.4 Partial Correlation: Deconfounding Age and Volume

Direct correlations: Volume $\rightarrow$ MAE ($\rho = -0.38$), Age $\rightarrow$ MAE ($\rho = -0.41$). Partial correlations: Volume (controlling age) $\rho_{\text{partial}} = -0.22$, Age (controlling volume) $\rho_{\text{partial}} = -0.08$. When controlling for volume, age effect nearly disappears, suggesting volume is primary driver.

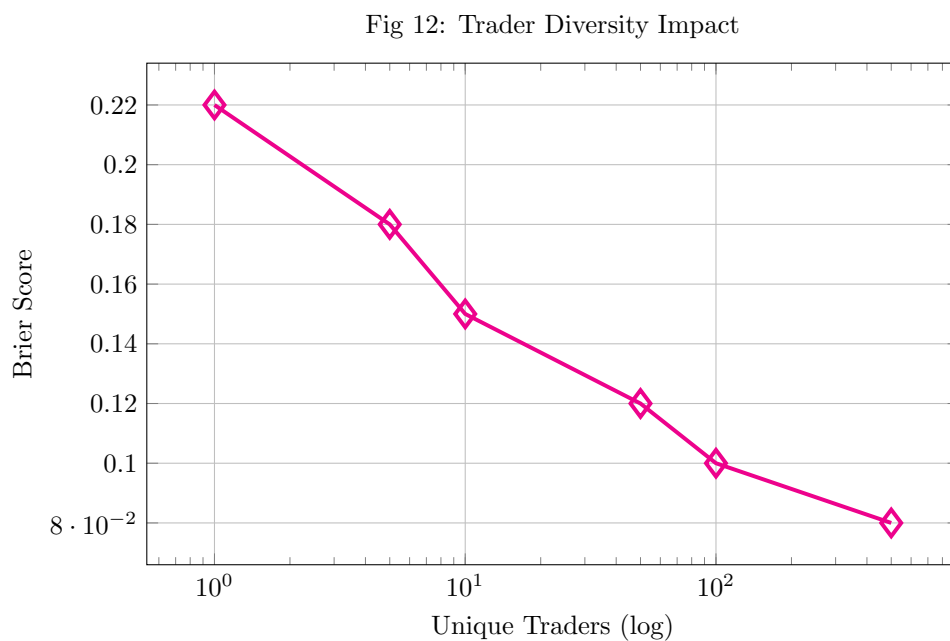


Figure 9: Wisdom of Crowds: 1 trader (0.22 Brier) $\rightarrow$ 500 traders (0.08 Brier).

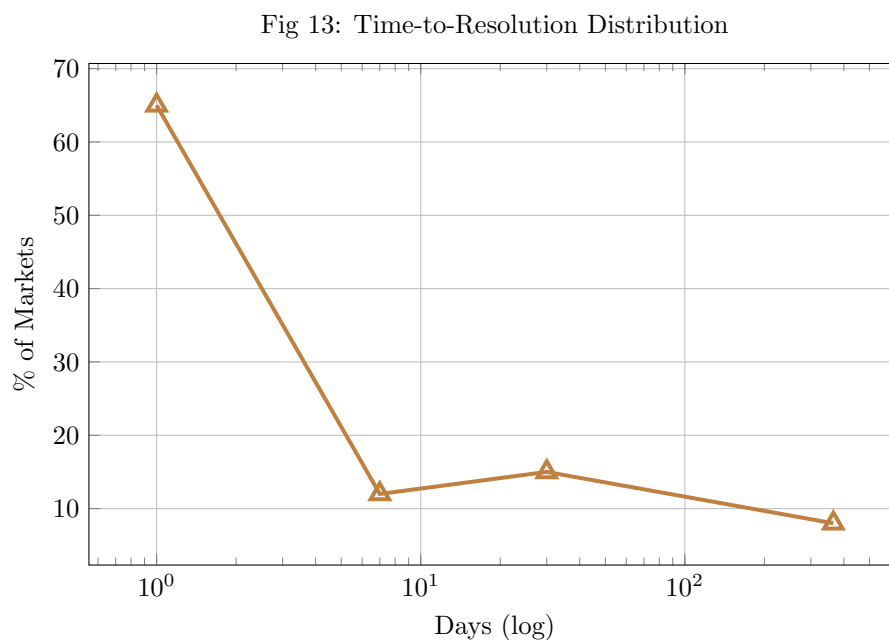


Figure 10: 65% resolve within 7 days; 8% exceed 365 days. Short horizons limit information accumulation.

Fig 14: Within-Market Lifecycle

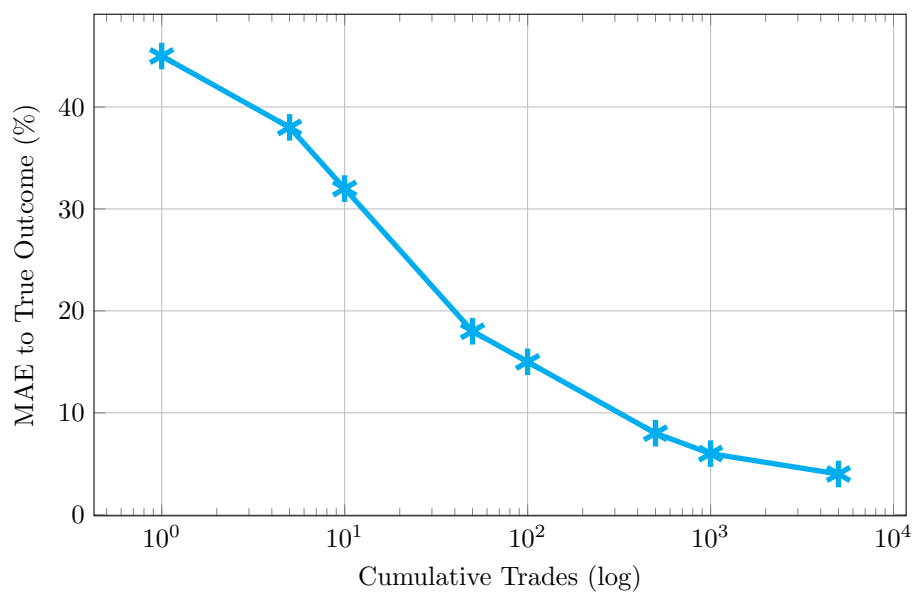


Figure 11: Early prices (1–5 trades) show 45% MAE; by 1,000 trades, 6% MAE. Sharp lifecycle convergence.

Fig 15: Volume Effect Robustness

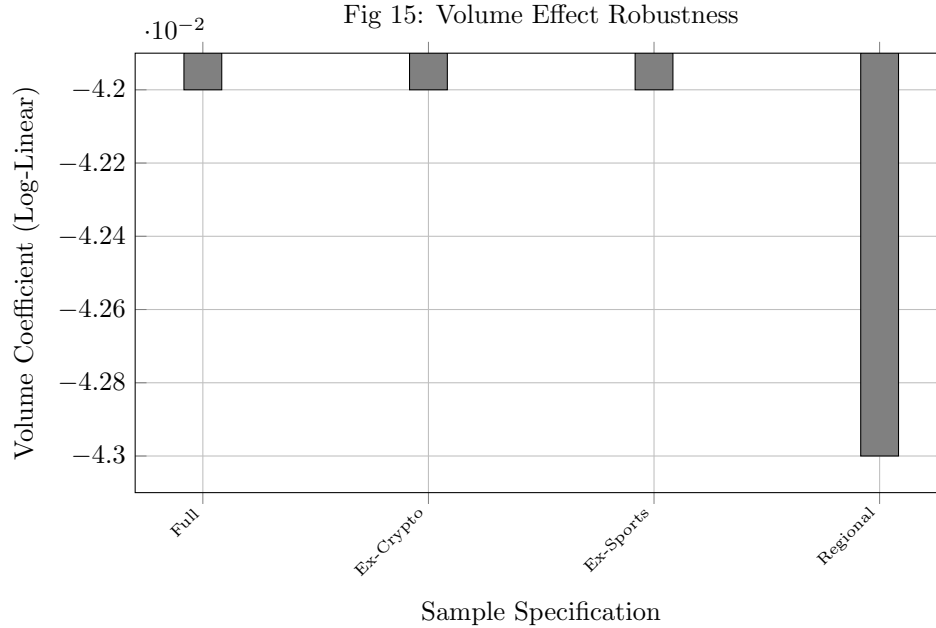


Figure 12: Volume coefficient stable (-0.042 to -0.043) across exclusions. Robust effect.